

Work Stream 1: Augmented Navier-Stokes & Thermodynamic Admissibility

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1 Introduction

Following discussions on [r/FluidMechanics](#), a consensus emerged: while the Lean 4 proof of finite-time blowup is syntactically flawless, it highlights the need for an augmented admissibility condition in 3D Navier-Stokes, analogous to the Lax entropy condition for the 1D inviscid Burgers equation.

2 The Admissibility Condition as a Model-Validity Condition

The Lax entropy condition selects, among weak solutions of Burgers' equation, the one that is the vanishing-viscosity limit: it encodes neglected physics (viscosity). The analogue for incompressible Navier-Stokes is a condition expressing the physics the model neglects, namely compressibility and molecular discreteness. A vortex of velocity U and size L has vorticity $|\omega| \sim U/L$; the model is being used within its constitutive assumptions only while $U \lesssim c$ (Mach number below one) and $L \gtrsim \lambda \sim \nu/c$ (Knudsen number below one; λ is the mean free path of a gas or the intermolecular spacing of a liquid, both of order ν/c). Together these give a *local* bound on vorticity:

$$\sup_{t \in [0, T)} \|\omega(\cdot, t)\|_{L^\infty} \lesssim \frac{c^2}{\nu}, \quad \text{equivalently} \quad \varepsilon = \nu |\omega|^2 \lesssim \frac{c^4}{\nu}. \quad (1)$$

For water at 300 K, $c^2/\nu \approx 2.2 \times 10^{12} \text{ s}^{-1}$; for air, $\approx 7.5 \times 10^9 \text{ s}^{-1}$. An earlier version of this work stream proposed instead a bound on the *global* integrated enstrophy $\int |\omega|^2 dx$ with a constant $\Omega_{\max} = 1.13 \times 10^{13} \text{ s}^{-2}$; that constant had no derivation, and a global L^2 bound on vorticity scales with the volume of fluid considered, so it does not express a material limit. Condition (1) is the form we now propose.

3 Why Admissible Solutions Do Not Blow Up, and Where the OpenAI Solution Exits

If a smooth solution satisfies (1) on $[0, T)$, then $\int_0^T \|\omega(t)\|_{L^\infty} dt \leq T c^2/\nu < \infty$, and the Beale–Kato–Majda criterion, valid for Navier–Stokes as well as Euler, extends the solution smoothly past T . So “admissible on $[0, T)$ implies no blow-up at T ” is a theorem whose only input is Definition (1) — not a postulate that excludes blow-up by fiat. Conversely, any finite-time singularity must first leave the admissible set. For the OpenAI construction the core has $\ell_r \simeq \sqrt{\nu t}$ and $u \simeq \sqrt{\nu/t}$ (up to $\tau^{\pm h}$, $h < 1/100$), so $|\omega| \simeq u/\ell_r \simeq 1/t$ reaches c^2/ν when ℓ_r reaches $\ell_* = \nu/c$, about 0.4 ps before the singularity in water (nanometre core), independently of the initial size of the vortex. The blow-up is thus a statement about the model continued past the scale ℓ_* where the fluid it describes would already be compressible, rarefied and heated by $\Delta T = u^2/c_p$.

4 Lean 4 Formalization

The accompanying `ThermodynamicAdmissibility.lean` file encodes the earlier global-entropy version and does not currently prove its main theorem, `admissible_no_blowup`, which has `sorry` in place of a proof body. The honest formal target is the BKM-based statement of the previous section — “ $\sup_t \|\omega\|_\infty \leq c^2/\nu$ on $[0, T)$ implies regularity on $[0, T]$ ” — stated against OpenAI’s own `CandidateProperties`; a genuine formalization requires the BKM theorem itself, which is not in Mathlib. This work stream should therefore be read as a proposed research direction with a well-defined formal target, not as a proven theorem.

5 Acknowledgments

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